A condition can force every functional below k to fail being a long alpha-decreasing sequence

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Editorial note. The counted unit is the article's own Lemma lem:coh-wf-forcing-wf at source lines 522-524 together with the proof environment that begins on the next line (525-543); the pairing is the ordinary adjacent one.

Editorial note. The retained context at source lines 480-491 is the author's own account of conditions, of compatibility of a string with a condition and of the two forcing clauses for the existential and universal jump formulas; the retained context at source line 505 is his own explanation of the relation forcing a decided initial segment of the jump.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, his natural-operation macros at source lines 104-107 and his quotation macro at source line 116. The LS2 font-encoding block at source lines 93-99, which declares \smashtimes through the stix fonts, is not retained because the host TeX Live has no stix directory and because the author never uses that symbol: source line 105 defines \ntimes as \dottimes and source line 110 shows the alternative already commented out. mathabx (source line 103) is retained because \dotplus and \dottimes come from it. The cleveref load at source line 6 and the inputenc load with the utf8x option at source line 8 are not retained, for the assembler's own cleveref machinery and the forced xelatex engine respectively. Nothing was renamed, redefined or added.

Editorial note. No retained line contains a citation or a cross-reference, so the material carries no bibliography and needs no navigation pointer.

Editorial note. The article's conservation theorems, its lemma lem:shore-blocking-wf, its lemma lem:wf-product, its lemma lem:coh-wf-forcing-jump and its proposition prop:coh-wf are not counted here and are not used as premises.

Allowed standard prerequisites (editorial).

- Conditions of the Cohen forcing used for COH. A condition is a pair (sigma, rho) with sigma a finite binary string and rho an M-finite approximation of the sequence of sets to be made cohesive, and it may be read

as a Mathias condition; a string tau is compatible with (sigma, rho) when sigma is an initial segment of tau and the difference tau minus sigma is contained in the members of the approximation; the article's own account is retained at source lines 480-483.. Fixes the two objects of the counted lemma, namely a condition (sigma,rho) and the compatibility relation over which the recursive construction runs.

- The forcing relation for the existential and universal jump formulas. A condition forces the existential statement that some value of the functional on the generic is defined when a witness and a bound smaller than the length of sigma exist, and it forces the universal statement that all values diverge when every string compatible with the condition fails to converge anywhere; the article's own definition is retained at source lines 485-491, and its explanation of forcing a decided initial segment of the jump at source line 505.. The counted lemma concludes a forcing statement about the generics G and Y, so the exact meaning of the forcing relation and of compatibility must be available; the proof uses the two clauses in its contradiction argument and its final transfer from conditions to generics.

- Theories Sigma^0_2 collection, Ramsey's theorem for singletons and well-foundedness. The proof uses that a model M satisfying the collection scheme for Sigma^0_2 formulas satisfies the Ramsey theorem for singletons with k colours for every k, and that M satisfying WF(alpha) means that M has no M-infinite sequence of ordinals below alpha that decreases at every step; the article states both notations and uses them throughout.. The proof obtains an M-infinite homogeneous set for the colours given by the indices below k from the collection scheme, and the contradiction is the existence of an M-infinite alpha-decreasing sequence, which WF(alpha) forbids.

One can think of a condition (σ, ρ) as a Mathias condition $(\sigma, \vec{R}_\rho)$. Requiring that $\rho \prec Q$ ensures that $\vec{R}_\rho$ is M-infinite. When increasing the length of ρ, we make progress in cohesiveness. Note that being a condition is a $\Delta^0_1(Q)$ predicate, as it simply requires checking that $\sigma \prec Q$, while the extension relation is $\Delta^0_1(Y)$. In particular, both are $\Delta^0_1(Q \oplus Y)$, so $\Delta^0_1(P)$.

A sequence τ is said to be *compatible* with a condition (σ, ρ) if $\sigma \prec \tau$ and $\tau - \sigma \subseteq \vec{R}_\rho$.

Definition 0.1. *Let (σ, ρ) be a condition, and Φ_e be a Turing functional.*

1. $(\sigma, \rho) \Vdash \exists x \Phi_e^{G \oplus Y}(x) \downarrow$ *if there is some* $x, t < |\sigma|$ *such that* $\Phi_e^{\sigma \oplus Y}(x)[t] \downarrow$
2. $(\sigma, \rho) \Vdash \forall x \Phi_e^{G \oplus Y}(x) \uparrow$ *if for every* τ *compatible with* (σ, ρ) *and every* $x \in M$, $\Phi_e^{\tau \oplus Y}(x) \uparrow$.

We will need the following two technical lemmas, which are Shore blocking arguments for satisfying the two kind of requirements, namely, deciding the jump and preserving $\mathsf{WF}(\alpha)$. In what follows, $(\tau, \rho) \Vdash (G \oplus Y)' \upharpoonright_k = \sigma'$ means

that for every $e < |\sigma'|$, if $\sigma'(e) = 1$ then $(\tau, \rho) \Vdash \Phi_e^{G \oplus Y}(e) \downarrow$ and if $\sigma'(e) = 0$, then $(\tau, \rho) \Vdash \Phi_e^{G \oplus Y}(e) \uparrow$. Thus, this relation is $\Pi^0_1(Y)$.

Lemma 0.2. *lemma/lem:coh-wf-forcing-wf Let (σ, ρ) be a condition. For every $k \in M$, there exists some $n \in M$ an extension $(\tau, \rho) \leqslant (\sigma, \rho)$ forcing $\Phi_e^{G \oplus Y}$ not to be an α-decreasing sequence of length more than n for every $e < k$.*

Proof. By twisting the Turing functionals, we can assume that for every $e, n \in M$, if $\Phi_e^\sigma(n) \downarrow$, then

- $\Phi_e^\sigma(m) \downarrow$ for every $m < n$
- $\Phi_e^\sigma(0), \Phi_e^\sigma(1), \ldots, \Phi_e^\sigma(m)$ is a strictly α-decreasing sequence

Suppose the lemma does not hold. Then in particular, for every M-finite string τ compatible with (σ, ρ) and every $n \in M$, there is some $\tau_1 \succeq \tau$ compatible with (σ, ρ) and some $e < k$ such that $\Phi^{\tau_1 \oplus Y}(n) \downarrow$.

Build Y-recursively an infinite sequence $\tau_0 \preceq \tau_1 \preceq \tau_2 \preceq \ldots$ such that for every n,

- τ_n is compatible with (σ, ρ)
- $\Phi_e^{\tau_n \oplus Y}(n) \downarrow$ for some $e < k$ (which depends on n)

Since $\mathcal{M} \models \mathsf{B}\Sigma^0_2$, $\mathcal{M} \models \forall k \mathsf{RT}^1_k$, so there is an M-infinite set $Z \in \mathcal{M}$ and some $e < k$ such that for every $n \in Z$, $\Phi_e^{\tau_n \oplus Y}(n) \downarrow$.

Then, letting $f(n) = \Phi_e^{\tau_n \oplus Y}(n)$, we have an M-infinite α-decreasing sequence, contradicting $\mathcal{M} \models \mathsf{WF}(\alpha)$. □